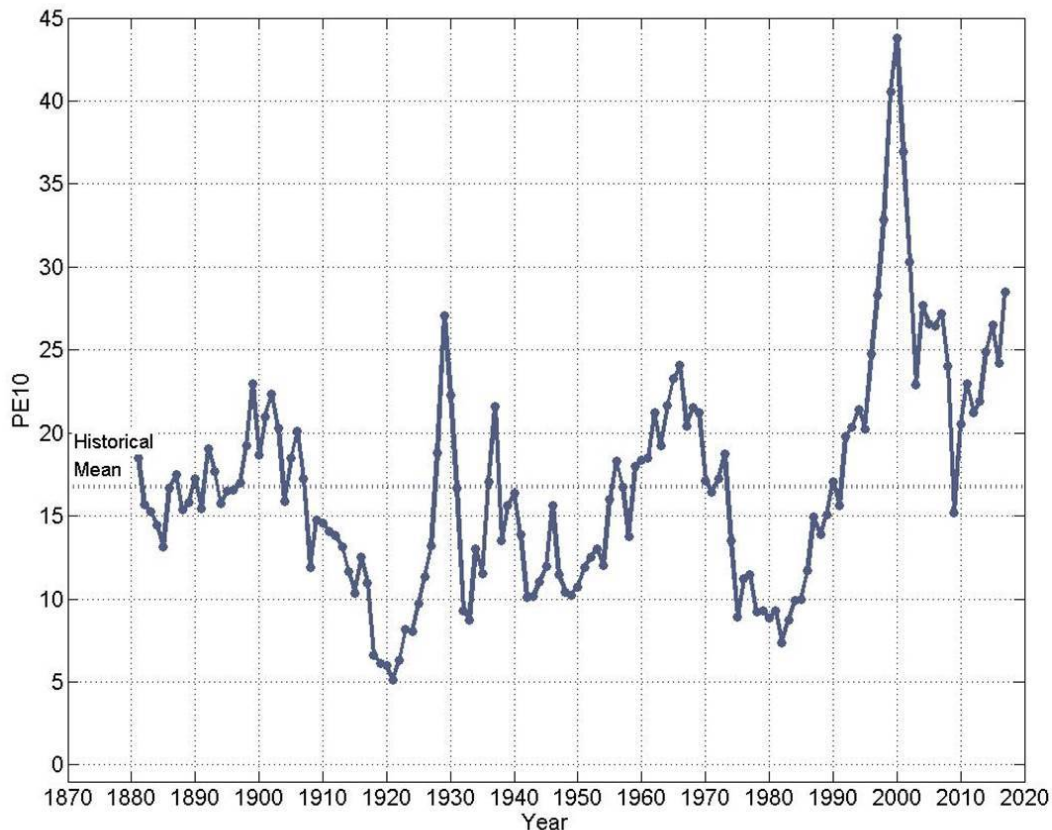


the past twenty years, relative to the period until 1987 for which we know the outcomes from simulated thirty-year retirements.

*Exhibit 5.4*

*Robert Shiller's Cyclically Adjusted Price-Earnings Ratio (PE10)*

*In January of Each Year, 1880–2017*



Source: Robert Shiller's data (<http://www.econ.yale.edu/~shiller/data.htm>).

Returns on bonds, meanwhile, depend on the initial bond yield and subsequent yield changes. Mathematically, if interest rates stay the same, then current interest rates will reflect the subsequent return on bonds. Low bond yields will tend to translate into lower returns due to less income and the heightened interest rate risk associated with capital losses if interest rates rise. This relationship is very tight. Exhibit 5.5 shows that recent retirees have been dealing with low interest rates. We do not have much experience